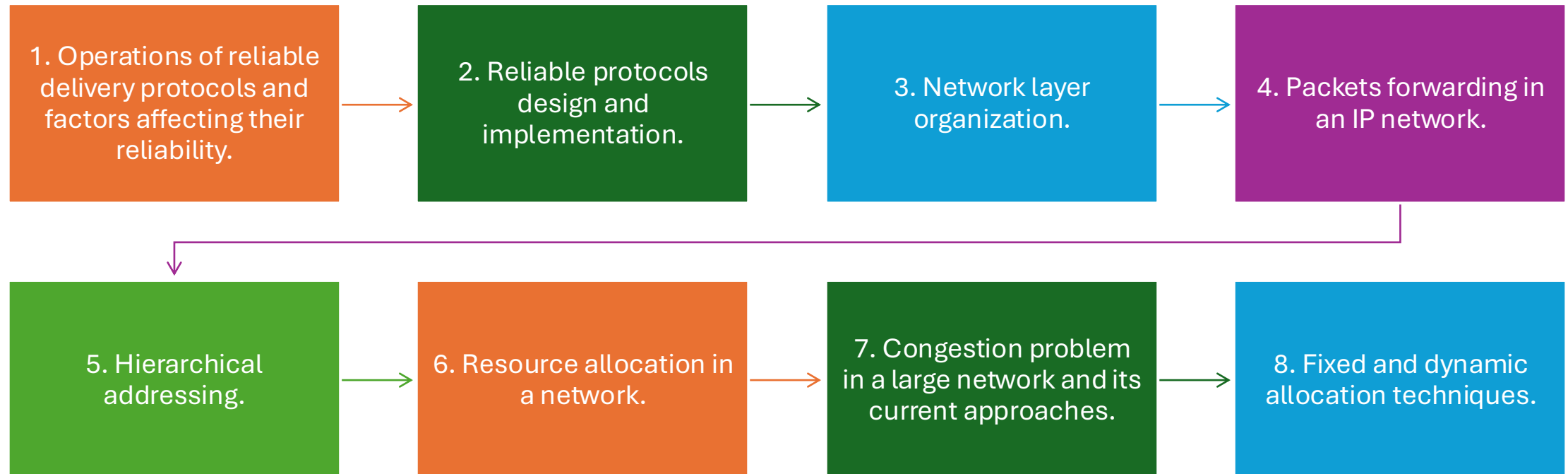


Reliable Data Delivery, Routing and Forwarding, and Resource Allocation

By : HoD - Dr. Mashari Huwaitem Al-Julaid

Topics to be covered



1 . Operations of reliable delivery protocols and factors affecting their reliability.

- Reliable delivery protocols are communication protocols designed to ensure that data sent from one device reaches the destination correctly, completely, and in the right order — even if the network is unreliable.
- These protocols detect and fix problems like:
 - Lost packets
 - Duplicated packets
 - Out-of-order packets
 - Transmission errors
- The most common reliable protocol in networking is TCP (Transmission Control Protocol).

- Reliable delivery protocols use several mechanisms to ensure reliability:

Operation	Description	Example
Acknowledgment (ACK)	The receiver confirms successful receipt of packets.	TCP sends ACKs after receiving data.
Retransmission	Lost packets are resent after a timeout if ACK is not received.	TCP's retransmission timer (RTO).
Sequence Numbers	Each packet is numbered to detect duplicates and maintain order.	TCP segments have sequence numbers.
Error Detection	Uses checksums to verify that data isn't corrupted during transmission.	TCP uses 16-bit checksum.
Flow Control	Prevents sender from overwhelming receiver with too much data.	TCP sliding window mechanism.
Congestion Control	Adjusts data rate based on network congestion.	TCP slow start, congestion avoidance.

• Factors Affecting Reliability

Topic# Operations of reliable delivery protocols and factors affecting their reliability

Factor	Effect on Reliability
Network Congestion	Causes packet loss and delay, reducing reliability.
Transmission Errors	Bit errors from noise or interference can corrupt data.
Packet Loss	Lost packets require retransmission, increasing delay.
High Latency	Slows acknowledgments and affects retransmission timing.
Routing Changes	Dynamic routing may reorder or drop packets temporarily.
Hardware/Software Faults	Malfunctioning devices or buggy firmware can cause data loss.

2. Reliable protocols design and implementation.



Reliable protocols design and implementation refers to the process by which network engineers and protocol developers create and build communication mechanisms that ensure data is transferred accurately, in sequence, without loss or duplication, even over networks that may drop, delay, or corrupt packets.



The goal is to make unreliable networks (like the Internet) behave as if they are perfectly reliable for the user.

Design of reliable protocols

Design Element	Purpose / Function	Example
Connection Establishment	Create a stable communication session between sender and receiver.	TCP's 3-way handshake.
Sequencing	Assign numbers to packets to keep data in correct order.	TCP sequence numbers.
Acknowledgment System	Confirm successful delivery of packets.	ACKs or cumulative ACKs.
Retransmission Strategy	Resend lost or unacknowledged packets.	Timeout or fast retransmit.
Flow Control	Balance data rate so the receiver isn't overloaded.	TCP sliding window.
Congestion Control	Prevent overload in the network.	Slow start, congestion avoidance.
Error Detection	Detect corruption in data.	Checksums.

Implementation of reliable protocols

Implementation Aspect	Description	Example
Coding in network stack	Writing the protocol logic in operating systems or routers.	TCP implemented in Linux kernel.
Timer Management	Handling retransmission timers efficiently.	TCP's adaptive timeout algorithm.
Buffer Management	Managing memory for unacknowledged data.	Sender and receiver buffers in TCP.
State Machine Control	Tracking connection states (established, closed, etc.).	TCP finite state machine.
Interfacing with Applications	Providing sockets or APIs for apps to use the protocol.	<code>socket ()</code> function in programming.

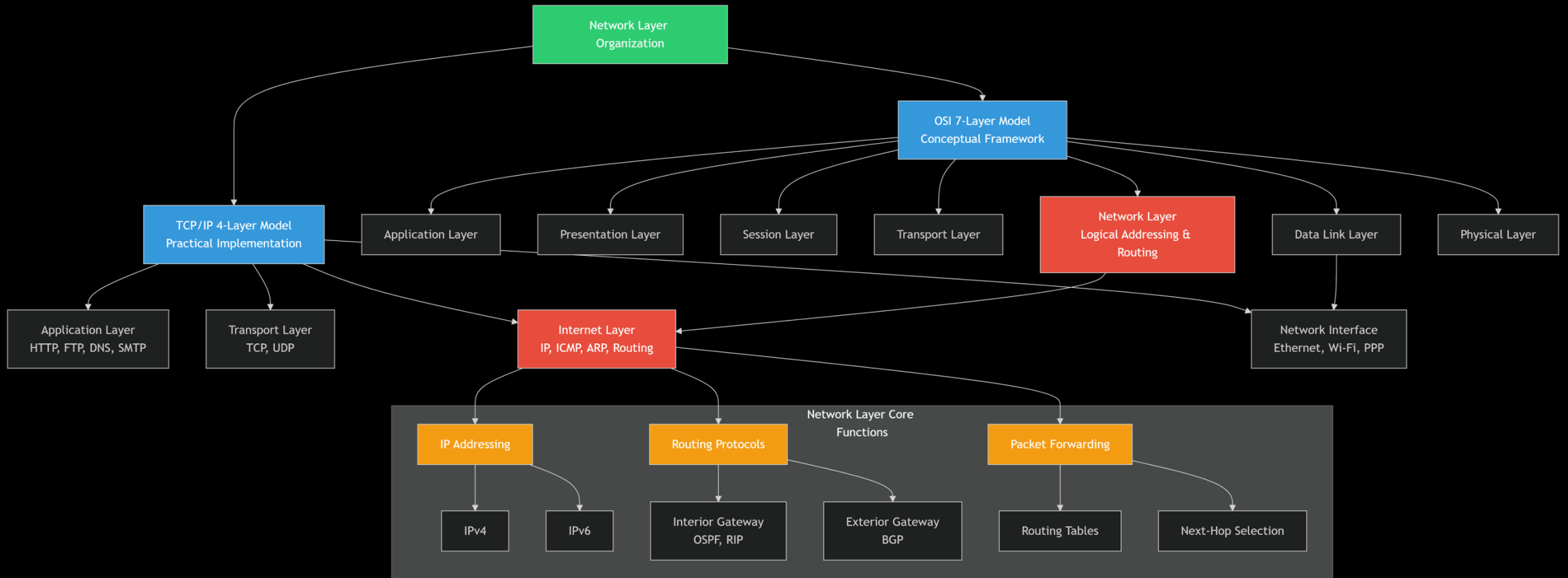
Topic# Reliable protocols design and implementation.

3. Network layer organization.

- Network Layer Organization refers to the internal structure, components, and functions of the Network Layer (Layer 3 in the OSI model or the Internet Layer in the TCP/IP model) that collectively enable data communication between devices across multiple, interconnected networks — forming what we know as the Internet.
- *In simple terms,*
- *It describes how the Network Layer is designed and arranged to handle logical addressing, routing, and forwarding of packets so that data can travel efficiently and correctly from a source device in one network to a destination device in another network — even if they are thousands of kilometers apart.*

Main purpose of Network Layer organization

- The main purpose of the network layer organization is to enable efficient, reliable, and scalable communication of data across multiple interconnected networks — ensuring that information can travel from the source device to the destination device, even if they are located on different networks or across the globe
- *In Technical Terms*
- *The organization of the network layer defines how logical addressing, routing, forwarding, and control mechanisms work together to manage data transmission across heterogeneous networks. It acts as a bridge between the transport layer (end-to-end communication) and the data link layer (physical network transmission).*



A hierarchical diagram of network layer organization

4. Packet forwarding in an IP network.



Packet forwarding refers to the process by which data packets are transferred from one network device to another — usually from a source computer to a destination computer — through a series of



When a device sends data, it's divided into smaller units called packets. Each packet carries:



Source IP address (where it came from),



Destination IP address (where it's going), and



Payload (the actual data).



Routers along the path read the destination IP address and use routing tables to decide the best next hop to move the packet closer to its destination. This process continues until the packet reaches the

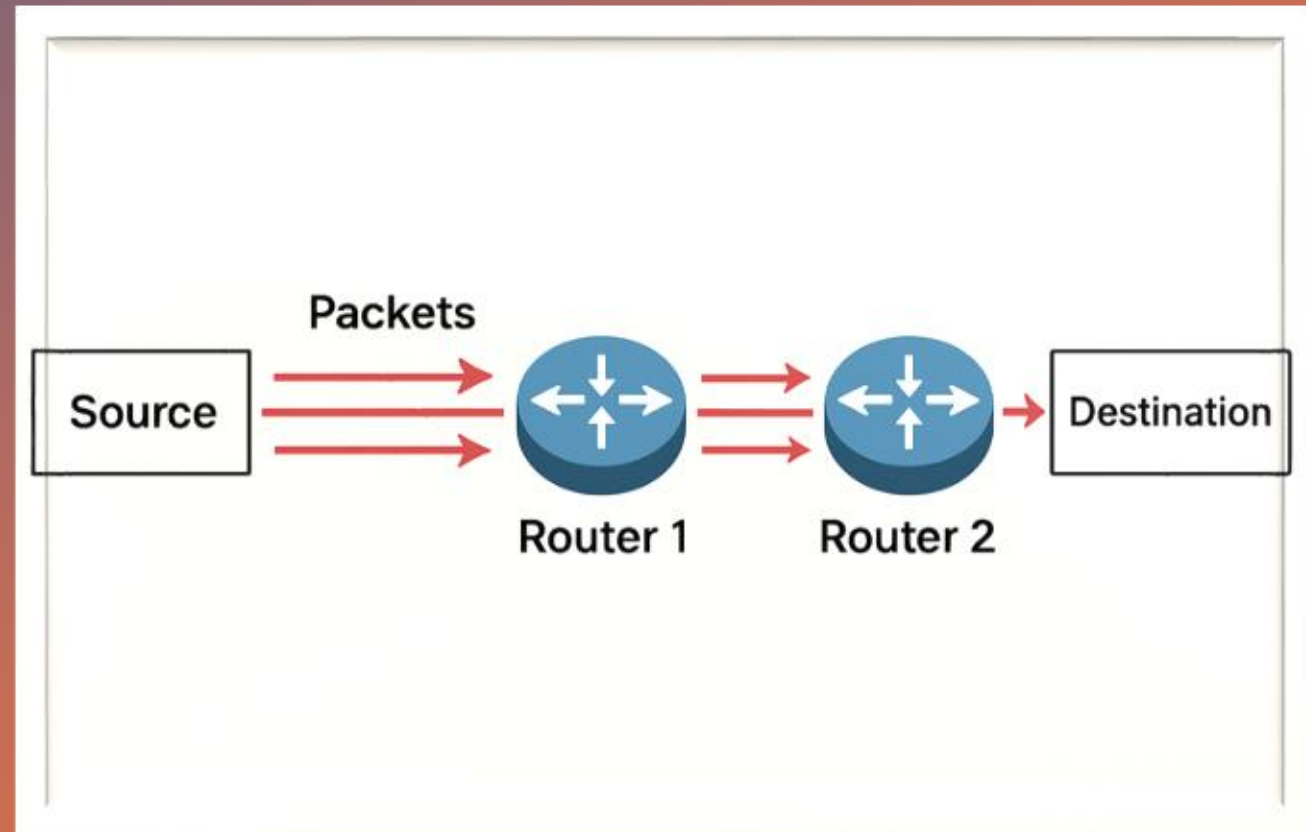
Packets forwarding in an IP network.

- Topic# Network layer organization.

- Packet forwarding is the core operational mechanism of the Internet Protocol (IP), responsible for moving packets from a source host to a destination host across multiple interconnected networks.
- Each device, usually a router, examines the destination IP address in the packet's header and uses its routing table to determine the optimal next hop based on metrics like hop count, cost, and network topology. Routers perform layer 3 processing, which may include decapsulation, header checksum verification, TTL (Time-To-Live) decrement, and re-encapsulation before transmitting the packet to the next router or the final destination.
- Forwarding decisions can follow different strategies — such as longest prefix matching, equal-cost multipath routing (ECMP), or policy-based routing — to optimize efficiency, reliability, and load distribution.
- Ultimately, packet forwarding ensures end-to-end connectivity, enabling data to traverse multiple autonomous systems while maintaining logical addressing independence from the underlying physical network.

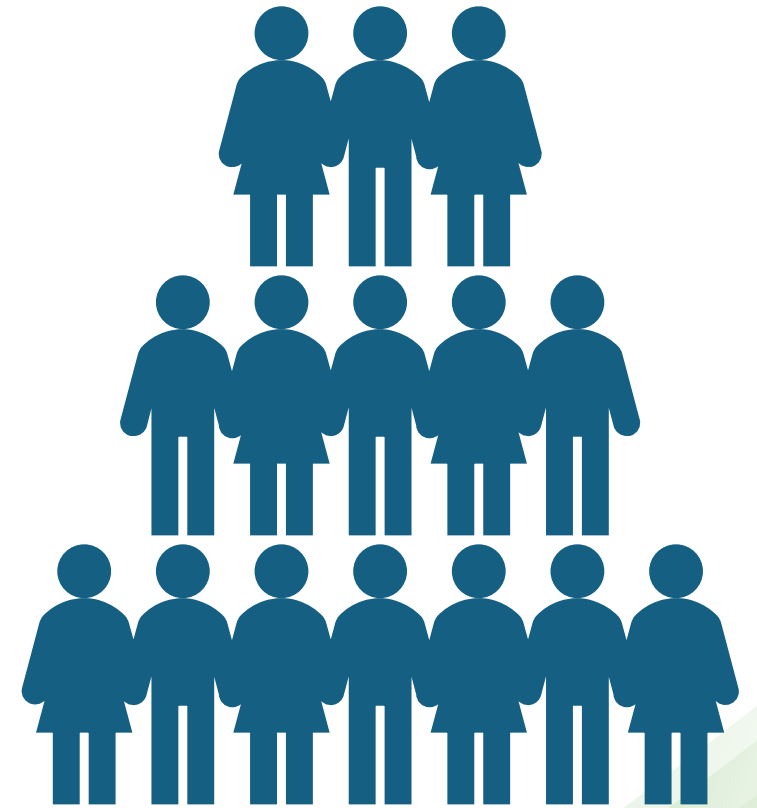
This is how packets travel across multiple routers in an IP network

Topic# Packet forwarding in a n IP network .



5. Hierarchical addressing.

- **Hierarchical addressing** refers to the **structured way of assigning network addresses** so that they are **organized in multiple levels (or layers)** — similar to how postal addresses are arranged (country → city → street → house number).
- In computer networks, this helps in **efficient routing, management, and scalability** of network addresses across large systems like the Internet.



Here's how hierarchy works:

Level	Part of Address	Meaning
Network Part	192.168	Identifies the organization or local network.
Subnet Part	10	Identifies a smaller subdivision of that network.
Host Part	25	Identifies the specific device or computer.

Why Hierarchical Addressing Is Important

Benefit	Explanation
Efficient Routing	Routers can make decisions based on network prefixes instead of every individual IP.
Scalability	Easier to manage billions of devices under organized address blocks.
Address Aggregation	Many small networks can be represented by a single large prefix (CIDR).
Simplified Management	Large organizations can divide address spaces logically (by department, branch, etc.).

6. Resource allocation in a network.

- Resource allocation in a network refers to the process of distributing and managing network resources — such as bandwidth, IP addresses, buffer space, and processing power — among different devices, users, or applications to ensure efficient and fair communication.
- It ensures that all users get the required network performance (speed, quality, and reliability) while avoiding congestion or unfair use of resources.

Resource Allocation in a Network – Summary Table

Resource Type	Common Allocation Method	Goal / Purpose
Bandwidth	Dynamic or Priority-based (e.g., QoS scheduling)	Ensure fair sharing and smooth data transmission for all users.
Buffer / Queue Space	Scheduling algorithms (FIFO, WFQ, Round Robin)	Avoid congestion and packet loss during heavy traffic.
IP Addresses	Static or Dynamic (via DHCP, subnetting)	Efficient and conflict-free identification of hosts in a network.
Channel Time (Wireless)	Time-slot or contention-based allocation	Reduce interference and maximize wireless throughput.
CPU / Memory (Routers, Servers)	Dynamic resource allocation	Improve performance and scalability during variable loads.

7. Congestion problem in a large network and its current approaches.

- Network congestion occurs when data traffic surpasses available capacity, causing delays and packet loss. Primary causes include excessive data volume, limited bandwidth, and buffer overflows. Effects manifest as high latency, packet loss, reduced throughput, and degraded service quality. Current mitigation approaches employ TCP rate control, queue management algorithms like RED, and traffic shaping techniques. Additional strategies include load balancing across paths and QoS prioritization for critical data. The overarching goal is to maintain efficient, fair, and smooth data flow. Effective congestion management is essential for optimal network performance and reliability.

Aspect	Short Explanation
Definition	Congestion occurs when data traffic exceeds network capacity, causing delay, packet loss, and reduced performance.
Causes	Too much data, limited bandwidth, buffer overflow, or inefficient routing.
Effects	High delay, packet loss, low throughput, poor QoS.
Current Approaches	- TCP Control: Adjusts sending rate. - Queue Management: RED, ECN. - Traffic Shaping: Leaky/Token Bucket. - Load Balancing: Distributes traffic. - QoS: Prioritizes important data.
Goal	Maintain smooth, fair, and efficient data flow across the network.

8. Fixed and dynamic allocation techniques.



Fixed and Dynamic Allocation

Techniques are the two core methods for managing how multiple devices share a single network channel. The fundamental problem they solve is **medium access control** - determining which device gets to transmit data and when, to prevent chaos and collisions.



Fixed Allocation divides the channel into predetermined, dedicated slots (by time or frequency) for each user. It's like giving every car a permanent lane on a highway - guaranteed but inefficient if lanes are empty.



Dynamic Allocation lets devices share the entire channel competitively, transmitting only when needed. It's like vehicles merging into active lanes - efficient but prone to collisions if unregulated.



THANK YOU